

常微分方程 复习课

1. 一阶线性常微分方程

$$y' + p(x)y = q(x) \quad y(x_0) = y_0 \quad (I)$$

解: (I) 两边同乘 $e^{\int_{x_0}^x p(t) dt}$ $\left(y e^{\int_{x_0}^x p(t) dt} \right)' = q(x) e^{\int_{x_0}^x p(t) dt}$

从 $x_0 \rightarrow x$ 积分 $y(x) e^{\int_{x_0}^x p(t) dt} - y(x_0) = \int_{x_0}^x q(t) e^{\int_{x_0}^t p(s) ds} dt$

$$\therefore y(x) = \frac{y(x_0) + \int_{x_0}^x q(t) e^{\int_{x_0}^t p(s) ds} dt}{e^{\int_{x_0}^x p(t) dt}}$$

$$y'' + \alpha y' = q(x)$$

2. 当 $p(x) = \alpha \neq 0$, 且 $\lim_{x \rightarrow +\infty} q(x) = A$ 存在且有界

(1) $\alpha > 0$ 时 证明 $\lim_{x \rightarrow +\infty} y(x)$ 存在且有界 并求出其极限

(2) $\alpha < 0$ 时 仅有一个初值 $y(x_0) = y_0$ 使得 $\lim_{x \rightarrow +\infty} y(x)$ 存在且有界

解: (1) 上题中含 $p(x) = \alpha$. $\Rightarrow e^{\int_{x_0}^x p(t) dt} = e^{\alpha(x-x_0)} \Rightarrow y(x) = \frac{y(x_0) + \int_{x_0}^x q(t) e^{\alpha(t-x_0)} dt}{e^{\alpha(x-x_0)}}$

当 $\alpha > 0$ 时 分母当 $x \rightarrow +\infty$ 时 $\rightarrow 0$

由 L'Hospital 法则得 $\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{q(x) e^{\alpha(x-x_0)}}{\alpha e^{\alpha(x-x_0)}} = \frac{A}{\alpha}$

(2) $\alpha < 0$ 时 分母 $\rightarrow 0$ 故分子必 $\rightarrow 0$. 即 $\lim_{x \rightarrow +\infty} \left[y(x_0) + \int_{x_0}^x q(t) e^{\alpha(t-x_0)} dt \right] = 0$

$$\Leftrightarrow y(x_0) + \int_{x_0}^{+\infty} q(x) e^{\alpha(x-x_0)} dx = 0 \Rightarrow y(x_0) = - \int_{x_0}^{+\infty} q(x) e^{\alpha(x-x_0)} dx$$

$$\Rightarrow y(x) = \frac{y(x_0) + \int_{x_0}^x q(t) e^{\alpha(t-x_0)} dt}{e^{\alpha(x-x_0)}} = \frac{\left[\int_{x_0}^x - \int_{x_0}^{+\infty} \right] q(t) e^{\alpha(t-x_0)} dt}{e^{\alpha(x-x_0)}} = \frac{- \int_x^{+\infty} q(t) e^{\alpha(t-x_0)} dt}{e^{\alpha(x-x_0)}}$$

这时 $\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{q(x) e^{\alpha(x-x_0)}}{\alpha e^{\alpha(x-x_0)}} = \frac{A}{\alpha}$

3. 解的唯一性

$y' = f(x, y)$ $y(x_0) = y_0$ 若 $f(x, y)$ 连续且满足 Lipschitz 条件

$|f(x, y_1) - f(x, y_2)| \leq L(x) |y_1 - y_2|$, 这里 $L(x)$ 是可积函数, 则 (I) 解唯一

证明: 设 (I) 有两个解 $y_1(x), y_2(x)$ $y_1(x) \neq y_2(x)$ $y_1(x_0) = y_2(x_0) = x_0$

$$\text{令 } V(x) = (y_2(x) - y_1(x))^2 \geq 0 \quad V(x_0) = 0$$

$$V'(x) = 2(y_2(x) - y_1(x))(y_2'(x) - y_1'(x)) = 2(y_2 - y_1)(f(x, y_2) - f(x, y_1))$$

$$\leq 2|y_2 - y_1| |f(x, y_2) - f(x, y_1)| \leq 2|y_2 - y_1| L(x) |y_2 - y_1| = 2L(x) |y_2 - y_1|^2$$

$$\Rightarrow V'(x) \leq 2L(x)V(x) \quad \text{两边同乘 } e^{-2\int_{x_0}^x L(t)dt}$$

$$\Rightarrow \left(V(x) e^{-2\int_{x_0}^x L(t)dt} \right)' \leq 0 \quad \text{两边从 } x_0 \rightarrow x (x > x_0) \text{ 积分}$$

$$V(x) e^{-2\int_{x_0}^x L(t)dt} - V(x_0) \leq 0 \Rightarrow V(x) \leq 0$$

又由于 $V(x) \geq 0$, 故 $V(x) \equiv 0, \forall x > x_0$ 四

对于 n 阶方程 (II) $y_1' = f_1, y_2' = f_2, \dots, y_n' = f_n, f_k = f_k(x, y_1, \dots, y_n) \quad k=1, 2, \dots, n$

若 f_k 满足 $|f_k(x, z_1, z_2, \dots, z_n) - f_k(x, y_1, y_2, \dots, y_n)| \leq L(x) \sum_{k=1}^n |z_k - y_k| \quad L(x)$ 可积

证明: (II) 满足初值条件 $y_k(x_0) = y_k^0$ 的解唯一 ($k=1, 2, \dots, n$)

提示: 设 (II) 有两个解 $z(x) = (z_1(x), \dots, z_n(x)) \quad y(x) = (y_1(x), \dots, y_n(x))$

$$\text{令 } V(x) = \|z(x) - y(x)\|^2 = \sum_{k=1}^n (z_k(x) - y_k(x))^2 \geq 0 \quad V(x_0) = 0$$

$$\Rightarrow V'(x) = 2 \sum_{k=1}^n (z_k - y_k)(z_k' - y_k') = 2 \sum_{k=1}^n (z_k - y_k) (f_k(x, z_1, \dots, z_n) - f_k(x, y_1, \dots, y_n))$$

$$\leq 2 \sum_{k=1}^n |z_k - y_k| |f_k(x, z_1, \dots, z_n) - f_k(x, y_1, \dots, y_n)|$$

$$\leq 2L(x) \left[\sum_{k=1}^n |z_k - y_k| \right] \left[\sum_{k=1}^n |z_k - y_k| \right] = 2L(x) \left[\sum_{k=1}^n |z_k - y_k| \right]^2$$

$$\text{有 } (\sum a_k b_k)^2 \leq \sum a_k^2 \sum b_k^2 \quad \text{Cauchy 不等式} \quad \text{令 } |z_k - y_k| = a_k \quad b_k = 1$$

$$\leq 2L(x) \left(\sum_{k=1}^n |z_k - y_k|^2 \right) \left(\sum_{k=1}^n 1 \right) = 2nL(x)V(x)$$

$$\Rightarrow V'(x) \leq 2nL(x)V(x) \quad \text{等式两边乘 } e^{-2n \int_{x_0}^x L(t) dt}$$

$$\Rightarrow \left(V(x) e^{-2n \int_{x_0}^x L(t) dt} \right)' \leq 0 \quad \text{从 } x_0 \rightarrow x \quad (x > x_0) \quad \text{积分}$$

$$V(x) e^{-2n \int_{x_0}^x L(t) dt} - V(x_0) \leq 0 \Rightarrow V(x) \leq 0$$

$$\text{同时 } V(x) \geq 0 \Rightarrow V(x) \equiv 0, \forall x > x_0. \quad \square$$

4. 齐次方程 $y' = f(\frac{y}{x})$. 令 $u = \frac{y}{x} \Rightarrow y = ux \Rightarrow y' = u'x + u$

$\Rightarrow u'x + u = f(u) \Rightarrow \frac{du}{dx} = \frac{f(u) - u}{x}$ ① $f(u) = u \Rightarrow \frac{du}{dx} = 0 \Rightarrow u = \text{常数}$

② $f(u) \neq u \Rightarrow \frac{du}{f(u) - u} = \frac{dx}{x} \Rightarrow \ln \left| \frac{x}{x_0} \right| = \int_{f(t)-t}^x \frac{dt}{f(t)-t}$

△ Bernoulli 方程 $y' + p(x)y = q(x)y^n \quad n \neq 0, 1$

$\Rightarrow y'y^{-n} + p(x)y^{1-n} = q(x) \quad \text{令 } z = y^{1-n} \quad z' = (1-n)y^{-n}y'$

两边同乘 $(1-n) \Rightarrow z' + (1-n)p(x)z = (1-n)q(x)$

上式化为一阶常微分方程 解出 $z(x)$. $\Rightarrow y(x) = z^{\frac{1}{1-n}}(x)$

5. 等角轨线 $k = \tan(\theta_2 - \theta_1) = \frac{\tan \theta_2 - \tan \theta_1}{1 + \tan \theta_2 \tan \theta_1}$

$\bar{y}' = \tan \theta_2 \quad \tan \theta_1 = y' = f(x, y) \Rightarrow k = \frac{\bar{y}' - y'}{1 + \bar{y}' y'}$

$\Rightarrow \bar{y}' - y' = k(1 + \bar{y}' y') \Rightarrow \bar{y}' = \frac{k + y'}{1 - k y'} = \frac{k + f(x, y)}{1 - k f(x, y)}$

其中, 当 $\theta_2 - \theta_1 = \frac{\pi}{2} \Rightarrow k = \pm \infty \Rightarrow \bar{y}' = \frac{1 + \frac{f}{k}}{\frac{k}{k} - f} \rightarrow -\frac{1}{f} \Rightarrow \bar{y}' = -\frac{1}{f(x, y)}$

例: 与直线族 $y = Cx$ 夹角为 θ_0 ($\tan \theta_0 = k$) 的曲线族

$y' = C = \frac{y}{x} \Rightarrow \bar{y}' = \frac{k + \frac{y}{x}}{1 - k \frac{y}{x}}$

$y' = \frac{k+u}{1-ku}, u = \frac{y}{x} \Rightarrow y = ux \Rightarrow y' = u'x + u \Rightarrow u'x + u = \frac{k+u}{1-ku}$

$\Rightarrow u'x = \frac{k+u}{1-ku} - u = \frac{k+u-u+ku^2}{1-ku} \Rightarrow u'x = \frac{k(1+u^2)}{1-ku} \Rightarrow \frac{1-ku}{1+u^2} du = \frac{dx}{x}$

两边积分 $\arctan u - \frac{k}{2} \ln(1+u^2) = \ln|x| + \ln C, C > 0$

将 $u = \frac{y}{x}$ 代回 $\Rightarrow \arctan \frac{y}{x} - \frac{k}{2} \ln \sqrt{1 + (\frac{y}{x})^2} = \ln|x| + \ln C \Rightarrow \sqrt{x^2 + y^2} = C' e^{\frac{1}{k} \arctan \frac{y}{x}} \quad C' > 0$

令 $\rho = \sqrt{x^2 + y^2} \quad \theta = \arctan \frac{y}{x} \Rightarrow \rho = C' e^{\frac{\theta}{k}} \quad \text{. 取 } \theta = \frac{\pi}{2} \Rightarrow \rho = C'$

5. Clairaut 方程 $y = xy' + f(y')$

通解: $y = Cx + f(C)$ 两边对 C 求导 $\Rightarrow 0 = x + f'(C) \Rightarrow x = -f'(C)$

$\Rightarrow y = -f'(C) + f(C)$ — 包络

6. 二阶常微分方程 $y'' + p(x)y' + q(x)y = 0$ (齐次) (*)

一定有两个线性无关解 $y_1(x), y_2(x)$

$\Rightarrow$ (*) 的解 $y(x) = C_1 y_1(x) + C_2 y_2(x)$. C_1, C_2 为任意常数

若已知 $\varphi(x)$ 是 (*) 的一个解, 则 $y_2(x) = \varphi(x) \cdot \int_{x_0}^x \frac{1}{\varphi^2(t)} e^{\int_{x_0}^t p(w)dw} dt$

$\Rightarrow y'' + p(x)y' + q(x)y = f(x)$ 的通解为

$$y(x) = C_1 \varphi(x) + C_2 y_2(x) + y^*(x), \quad y^*(x) = \int_{x_0}^x k(x,t) f(t) dt, \quad k(x,t) = \frac{\begin{vmatrix} \varphi(t) & \varphi(x) \\ y_2(t) & y_2(x) \end{vmatrix}}{\begin{vmatrix} \varphi(t) & y_2(t) \\ \varphi'(t) & y_2'(t) \end{vmatrix}}$$

7. Wronsky 行列式 (不要忘)

$$\text{令 } W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}, \quad W'(x) = \begin{vmatrix} y_1'(x) & y_2'(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} + \begin{vmatrix} y_1(x) & y_2(x) \\ y_1''(x) & y_2''(x) \end{vmatrix}$$

$$= \begin{vmatrix} y_1(x) & y_2(x) \\ -py_1' - qy_1 & -py_2' - qy_2 \end{vmatrix} = \begin{vmatrix} y_1 & y_2 \\ -py_1 & -py_2 \end{vmatrix} = -p(x)W(x) \Rightarrow W'(x) + p(x)W(x) = 0$$

两边同乘 $e^{\int_{x_0}^x p(t)dt}$

$\Rightarrow \left(W(x) e^{\int_{x_0}^x p(t)dt} \right)' = 0$ 从 $x_0 \rightarrow x$ 积分.

$$\Rightarrow W(x) e^{\int_{x_0}^x p(t)dt} - W(x_0) = 0 \Rightarrow W(x) = W(x_0) e^{-\int_{x_0}^x p(t)dt}$$

Liouville 公式

8. $y'' + py' + qy = f(x)$ p, q 是实常数

令 $y(x) = e^{\lambda x}$ 代 $\lambda \Rightarrow \lambda^2 + p\lambda + q = 0$ 特征方程 $\Delta = p^2 - 4q$

1/ $p^2 - 4q > 0 \Rightarrow$ 有两不等实根 $\lambda_1, \lambda_2, \lambda_1 \neq \lambda_2$

$y(x) = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} + y^*(x), C_1, C_2$ 是任意常数 $y^*(x) = \int_{x_0}^x K(x,t) f(t) dt$

$$K(x,t) = \frac{\begin{vmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} \\ e^{\lambda_1 x} & e^{\lambda_2 x} \end{vmatrix}}{\begin{vmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} \\ \lambda_1 e^{\lambda_1 t} & \lambda_2 e^{\lambda_2 t} \end{vmatrix}}$$

2/ $p^2 - 4q = 0 \Rightarrow$ 有一对重根 $\lambda_1 = \lambda_2 = \lambda = -\frac{p}{2}$

$y(x) = (C_1 + C_2 x) e^{\lambda x} + y^*(x), C_1, C_2$ 是任意常数 $y^*(x) = \int_{x_0}^x K(x,t) f(t) dt$

$$K(x,t) = \frac{\begin{vmatrix} e^{\lambda t} & t e^{\lambda t} \\ e^{\lambda x} & x e^{\lambda x} \end{vmatrix}}{\begin{vmatrix} e^{\lambda t} & t e^{\lambda t} \\ \lambda e^{\lambda t} & (1+t\lambda) e^{\lambda t} \end{vmatrix}}$$

3/ $p^2 - 4q < 0 \Rightarrow$ 有一对共轭复根 $\lambda_{1,2} = \alpha \pm i\beta, \alpha = -\frac{p}{2}, \beta = \frac{\sqrt{-\Delta}}{2} > 0$

这时 $y(x) = (C_1 \cos \beta x + C_2 \sin \beta x) + y^*(x), C_1, C_2$ 是任意常数 $y^*(x) = \int_{x_0}^x K(x,t) f(t) dt$

$$K(x,t) = \frac{\begin{vmatrix} e^{\alpha t} \cos \beta t & e^{\alpha t} \sin \beta t \\ e^{\alpha x} \cos \beta x & e^{\alpha x} \sin \beta x \end{vmatrix}}{\begin{vmatrix} e^{\alpha t} \cos \beta t & e^{\alpha t} \sin \beta t \\ e^{\alpha t} (-\beta \sin \beta t + \alpha \cos \beta t) & e^{\alpha t} (\beta \cos \beta t + \alpha \sin \beta t) \end{vmatrix}} = \frac{e^{\alpha(x-t)} \sin \beta(x-t)}{\beta}$$

当 $\alpha = 0, \beta = n, n$ 是正整数时 $y(x) = C_1 \cos nx + C_2 \sin nx + y^*(x)$

$$y^*(x) = \frac{\sin nx}{n} \int_{x_0}^x \cos t f(t) dt$$

Euler 方程 $x^2 y'' + xpy' + qy = f(x) \quad (x > 0)$

令 $x = e^t \Rightarrow t = \ln x$

$$\frac{d^2 y}{dt^2} + (p-1) \frac{dy}{dt} + qy = f(e^t)$$

$$\Delta = (p-1)^2 - 4q > 0.$$

$y(x) = C_1 x^{\lambda_1} + C_2 x^{\lambda_2} + y^*(\ln x)$, $\lambda_{1,2} = \frac{-(p-1)}{2} \pm \sqrt{\frac{\Delta}{2}}$, C_1, C_2 是常数

$$y^*(\ln x) = \int_{x_0}^x k(x,t) f(e^t) dt, \quad k(x,t) = \frac{\begin{vmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} \\ e^{\lambda_1 x} & e^{\lambda_2 x} \end{vmatrix}}{\begin{vmatrix} e^{\lambda_1 t} & e^{\lambda_2 t} \\ \lambda_1 e^{\lambda_1 t} & \lambda_2 e^{\lambda_2 t} \end{vmatrix}}$$

全微分方程 — 条件

$$P(x,y)dx + Q(x,y)dy = 0 \quad (I)$$

$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ 时 $\Rightarrow (I)$ 是全微分方程 $\Rightarrow u(x,y) = C$, C 是常数.

其中 $\frac{\partial u}{\partial x} = P(x,y)$ $\frac{\partial u}{\partial y} = Q(x,y)$

$$u(x,y) = \int_{x_0}^x P(t,y) dt + q(y)$$

$$\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \left(\int_{x_0}^x P(t,y) dt \right) + q'(y) = Q(x,y) \text{ 求得 } q'(y), \text{ 积分得 } q(y)$$

例1 $(2x \sin y + 3x^2 y) dx + (x^3 + x^2 \cos y + y^2) dy = 0$

$$\frac{\partial P}{\partial y} = 2x \cos y + 3x^2 \quad \frac{\partial Q}{\partial x} = 3x^2 + 2x \cos y \quad \text{故上式是全微分方程}$$

$$u(x,y) = \int_0^x (2t \sin y + 3t^2 y) dt + q(y) = x^2 \sin y + x^3 y + q(y)$$

$$\frac{\partial u}{\partial y} = x^2 \cos y + x^3 + q'(y) = x^3 + x^2 \cos y + y^2 \Rightarrow q'(y) = y^2 \Rightarrow q(y) = \frac{1}{3} y^3 + C, C \text{ 是常数.}$$

$$\Rightarrow u(x,y) = x^2 \sin y + x^3 y + \frac{1}{3} y^3 + C$$

例1 $(2x\sin y + 3x^2y)dx + (x^3 + x^2\cos y + y^2)dy = 0$

$$(2x\sin y dx + x^2\cos y dy) + (3x^2y dx + x^3 dy) + y^2 dy = 0$$

$$d(x^2\sin y) + d(x^3y) + d(\frac{1}{3}y^3) = 0$$

$$d(x^2\sin y + x^3y + \frac{1}{3}y^3) = 0$$

微积分常用公式

① $ydx + xdy = d(xy)$ ⑤ $\frac{-ydx + xdy}{x^2} = d(\frac{y}{x})$

② $\frac{ydx - xdy}{x^2 + y^2} = d(\arctan \frac{y}{x})$

③ $\frac{xdy - ydx}{x^2 + y^2} = d(\arctan \frac{y}{x}) = d\theta$

④ $\frac{ydx - xdy}{y^2} = d(\frac{x}{y})$

Hurwitz 定理

$\lambda^n + a_1\lambda^{n-1} + \dots + a_n = 0$ 特征方程, $a_1, \dots, a_n$ 是实数

$\operatorname{Re}(\lambda_k) < 0, k=1, 2, \dots, n$. 称该方程稳定

Hf =
$$\begin{vmatrix} a_1 & 1 & 0 & 0 & \dots & 0 \\ a_3 & a_2 & a_1 & 1 & \dots & 0 \\ a_5 & a_4 & a_3 & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{2n-1} & a_{2n-2} & a_{2n-3} & \dots & \dots & a_n \end{vmatrix} \quad a_k = 0, k \geq n+1 \text{ 时}$$

方程稳定 $\Leftrightarrow$ Hf 每一个主子式 $\Delta_k > 0$

$$\textcircled{1} \quad n=2 \quad \Delta_1 > 0 \Rightarrow a_1 > 0 \quad \Delta_2 > 0 \Rightarrow a_2 \Delta_1 > 0 \Rightarrow a_2 > 0$$

$$\textcircled{2} \quad n=3 \quad \Delta_1 > 0 \Rightarrow a_1 > 0 \quad \Delta_3 > 0 \Rightarrow a_3 \Delta_2 > 0 \Rightarrow a_3 > 0$$

$$\Delta_2 > 0 \Rightarrow a_1 a_2 > a_3$$

$$\textcircled{3} \quad n=4 \quad \Delta_1 > 0 \Rightarrow a_1 > 0 \quad \Delta_4 > 0 \Rightarrow a_4 \Delta_3 > 0 \Rightarrow a_4 > 0$$

$$\Delta_2 > 0 \Rightarrow a_1 a_2 > a_3 \quad \Delta_3 > 0 \Rightarrow a_1 a_2 a_3 > a_1^2 a_4 + a_3^2$$